V.L Memorial Public School

475 Vijay Nagar

CHAPTER-8-QUESTIONS-CLASS8TH-SCIENCE

Class 08 - Science (Curiosity)

Section A

1. Why might milk not be considered a pure substance in scientific terms? **[1]**

   a) It is a mixture of different components
   b) It contains only one type of particle
   c) It is naturally sourced
   d) It is white in color

2. Which statement correctly describes a native mineral? **[1]**

   a) A compound of elements
   b) A mixture of minerals
   c) A pure element
   d) A mixture of elements

3. Which gas produces a pop sound near a burning candle? **[1]**

   a) Hydrogen sulfide gas
   b) Hydrogen gas
   c) Oxygen gas
   d) Water vapour

4. Sugar dissolved evenly in water forms which type of mixture? **[1]**

   a) Layered mixture
   b) Solid mixture
   c) Uniform mixture
   d) Non-uniform mixture

5. What kind of substance is pure and has identical properties throughout? **[1]**

   a) Element
   b) Suspension
   c) Mixture
   d) Non-uniform mixture

6. Which material is highly porous and has a high absorbing capacity? **[1]**

   a) Pure iron
   b) Solid aluminium
   c) Liquid water
   d) Graphene aerogel

7. A solution of salt in water is an example of **[1]**

   a) Element
   b) Non-uniform mixture
   c) Compound
   d) Uniform mixture

8. Which property identifies a substance as pure in science? **[1]**

   a) It contains cheaper substances
   b) It contains several substances
   c) It contains one substance
   d) It contains poorer substances

9. What substance forms when iron and sulfur are heated together? **[1]**

   a) Iron sulfide
   b) Sodium chloride
   c) Hydrogen sulfide
   d) Iron chloride

10. Which alloy contains copper and zinc? **[1]**

a) Brass
b) Bronze
c) Stainless steel
d) Mishraloha

11. What is the ratio of hydrogen atoms to oxygen atoms in one molecule of water? **[1]**

a) 1 : 1
b) 2 : 1
c) 1 : 2
d) 3 : 1

12. Why are mixtures separated scientifically? **[1]**

a) For aesthetic purposes
b) To obtain pure substances
c) For better taste
d) To save space

13. A compound is formed when **[1]**

a) Two gases are combined physically
b) Two solids are mixed
c) One element changes state
d) Two or more elements combine chemically

14. What is a common misconception about pure substances? **[1]**

a) They taste better than mixtures
b) They can be easily found outdoors
c) They are always liquids
d) They are substances that are considered unadulterated by people

15. Which of the following has a variable composition? **[1]**

a) Mixture
b) Molecule
c) Compound
d) Element

16. How are the components of non-uniform mixtures usually arranged? **[1]**

a) Evenly distributed
b) Randomly scattered
c) Arranged in layers
d) Tightly packed

17. Why are chemists concerned with understanding how elements combine? **[1]**

a) To create lightweight materials
b) To provide faster computing
c) To invent life-saving medicines
d) To build stronger structures

18. Which of the following is not a metal native mineral? **[1]**

a) Sulfur
b) Copper
c) Gold
d) Silver

19. How many chemical elements are currently known? **[1]**

a) 108 elements
b) 128 elements
c) 138 elements
d) 118 elements

20. Which of the following is a compound mineral? **[1]**

a) Pyroxene
b) Sulfur

c) Copper
d) Gold

21. **Assertion (A):** Native minerals are composed of more than one element. **[1]**
Reason (R): Sulfur, carbon, and gold are examples of native minerals.

a) Both A and R are true and R is the correct explanation of A.
b) Both A and R are true but R is not the correct explanation of A.
c) A is true but R is false.
d) A is false but R is true.

22. **Assertion (A):** Cement is made from materials extracted from minerals. **[1]**
Reason (R): Calcite, quartz, alumina, and iron oxide are components of cement.

a) Both A and R are true and R is the correct explanation of A.
b) Both A and R are true but R is not the correct explanation of A.
c) A is true but R is false.
d) A is false but R is true.

23. **Assertion (A):** All elements are substances made of identical atoms. **[1]**
Reason (R): Elements cannot be separated into simpler substances by chemical means.

a) Both A and R are true and R is the correct explanation of A.
b) Both A and R are true but R is not the correct explanation of A.
c) A is true but R is false.
d) A is false but R is true.

24. **Assertion (A):** Mixtures can involve solids, liquids, and gases in various combinations. **[1]**
Reason (R): Each component of a mixture has distinct visibility in all mixtures.

a) Both A and R are true and R is the correct explanation of A.
b) Both A and R are true but R is not the correct explanation of A.
c) A is true but R is false.
d) A is false but R is true.

25. **Assertion (A):** A pure substance in science is not mixed with any other material. **[1]**
Reason (R): Adding substances to a product increases its purity according to scientific standards.

a) Both A and R are true and R is the correct explanation of A.
b) Both A and R are true but R is not the correct explanation of A.
c) A is true but R is false.
d) A is false but R is true.

26. **Assertion:** Air is a mixture. **[1]**
Reason: A mixture is formed when two or more substances are mixed, without undergoing any chemical change.

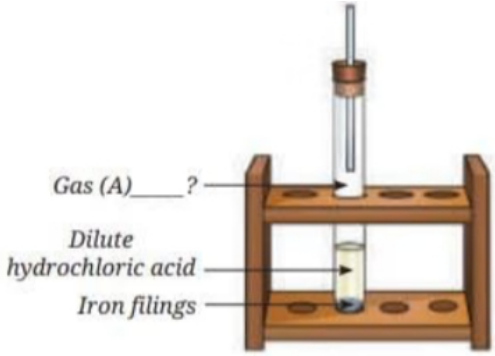
a) Both A and R are true and R is the correct explanation of A.
b) Both A and R are true but R is not the correct explanation of A.
c) A is true but R is false.
d) A is false but R is true.

27. **Assertion (A):** Stainless steel is an example of an alloy. **[1]**
Reason (R): Alloys are mixtures wherein the individual components can be seen separately with the naked eye.

a) Both A and R are true and R is the correct explanation of A.
b) Both A and R are true but R is not the correct explanation of A.

c) A is true but R is false. d) A is false but R is true.

28. Identify the two gases formed when water is electrolyzed in the presence of sulfuric acid. **[1]**
29. Explain why stainless steel is considered a uniform mixture. **[1]**
30. Why can't we separate hydrogen and oxygen from water using physical methods? **[1]**
31. Explain why water is classified as a compound and not a mixture. **[1]**
32. In what way can adulteration harm consumers? **[1]**
33. Analyse figure. Identify Gas A. Also, write the word equation of the chemical reaction. **[1]**



34. Describe a simple procedure to separate sugar into its constituent components. **[1]**

Section B

35. Why is sprout salad considered a non-uniform mixture? **[2]**
36. Explain how minerals are useful in everyday life, giving the examples of cement and talcum powder. **[2]**
37. How are most minerals different from native minerals? **[2]**
38. How are elements distributed among different physical states at room temperature? **[2]**
39. How can elements be combined to form a compound? **[2]**
40. Explain why adulteration is considered harmful to a product. **[2]**
41. How would our daily lives be changed if water were not a compound but a mixture of hydrogen and oxygen? **[2]**

Section C

42. Describe how the concept of mixtures can be observed in everyday life. **[3]**
43. Explain why certain mixtures appear uniform while others do not. **[3]**
44. Describe the properties and potential uses of graphene aerogel. **[3]**
45. State the properties of a solution. **[3]**
46. Define pure substances and differentiate between elements and compounds within pure substances. **[3]**
47. List two examples of alloys and describe what makes them mixtures. **[3]**
48. Describe how elements differ from compounds, highlighting their properties and examples. **[3]**

Section D

49. **Read the following text carefully and answer the questions that follow:** **[4]**

Pure substances are classified as elements and compounds. Elements, like hydrogen, oxygen, gold, and silver, consist of identical atoms and cannot be broken into simpler substances. Compounds, such as water, sugar, and iron sulfide, are formed when elements combine in fixed ratios, producing new substances with distinct properties. Unlike

mixtures, compounds cannot be separated by physical methods, and their chemical composition remains constant, regardless of the sample size.

Questions:

1. Give an example of a compound and explain why it is considered a compound. **(1)**
2. Explain why iron sulfide is classified as a compound. **(1)**
3. Differentiate between an element and a compound with examples. **(2)**

   OR

   Explain why compounds like water cannot be separated into elements by physical methods. **(2)**

50. **Read the following text carefully and answer the questions that follow:** **[4]**

Most rocks are mixtures of minerals, which can be elements or compounds. Native minerals, like gold and sulfur, are pure elements, while others, such as quartz and calcite, are compounds. Minerals are used in everyday materials like cement and talcum powder. Traditional Indian crafts like Dhokra art use metals to create sculptures. Elements and compounds form all matter, but phenomena like light, heat, electricity, and emotions are not matter.

Questions:

a. Give an example of a native mineral and explain why it is called native. **(1)**
b. Explain the difference between native minerals and other minerals. **(1)**
c. Describe the use of minerals in daily life and traditional crafts. **(2)**

   OR

   Why is it important to distinguish between matter and non-matter? **(2)**

51. **Read the following text carefully and answer the questions that follow:** **[4]**

In everyday life, 'pure' often means unadulterated, like milk or ghee without additives. Adulteration adds cheaper substances, reducing quality and health safety. In science, a pure substance contains only one type of particle and cannot be separated physically into other substances. Even products that appear pure may be mixtures. For example, milk or packed fruit juice are mixtures, while sugar and baking soda are considered pure substances.

Questions:

a. Give an example of a pure substance and explain why it is classified as pure in science. **(1)**
b. Define a pure substance in scientific terms. **(1)**
c. Explain the difference between 'pure' in everyday life and 'pure' in science. **(2)**

   OR

   Classify milk, baking soda, sugar, and soil as pure substances or mixtures and justify your answer. **(2)**

Section E

52. Describe the significance of water as a compound and its role in sustaining life and the environment. **[5]**
53. How does the purpose of separating mixtures differ in everyday life and in science? **[5]**
54. Evaluate the statement: Every pure substance is an element. **[5]**

55. Explain the following giving examples. **[5]**

a. saturated solution

b. pure substance

c. colloid

d. suspension

56. Write a method to separate different gases from air. **[5]**

Section F

57. **Fill in the blanks:** **[9]**

(a) A mixture contains more than _____ substance mixed in any proportion. **[1]**

(b) _____ are the building block of all substances. **[1]**

(c) Sodium and chlorine combine to form a substance known as _____. **[1]**

(d) In a compound, atoms are bonded together in a definite _____. **[1]**

(e) The smallest particles that make up all matter are called _____. **[1]**

(f) Two or more elements chemically combined form a _____. **[1]**

(g) _____ can be separated by physical methods such as filtration or evaporation. **[1]**

(h) Sugar and water form a _____ mixture. **[1]**

(i) Water is a compound made of hydrogen and oxygen in a _____ ratio. **[1]**

Section G

58. **State whether the given statement is True or False:** **[10]**

(a) 'Pure' in a scientific context means unadulterated products. **[1]**

(b) The air is a non-uniform mixture. **[1]**

(c) Elements cannot be broken down into simpler substances by chemical means. **[1]**

(d) The common usage of 'pure' and scientific usage of 'pure' have the same meaning. **[1]**

(e) Adulteration improves the quality of consumable products. **[1]**

(f) The smallest particle of an element is a molecule. **[1]**

(g) Evaporation is a physical method used to separate salt from water. **[1]**

(h) All metals and non-metals exist in the solid state at room temperature. **[1]**

(i) Brass is an example of an alloy. **[1]**

(j) Sugar dissolved in water forms a uniform mixture. **[1]**